

Exercise #1

Consider the RRPR robot in Fig. 1, where all relevant kinematic and dynamic parameters are also shown. The robot moves in a vertical plane. Compute the inertia matrix $M(q)$ and the gravity vector $g(q)$ in the Lagrangian dynamic model of this robot. Provide a linear factorization of each term, $M(q)\ddot{q} = Y_M(q, \dot{q})a_M$ and $g(q) = Y_g(q)a_g$, introducing dynamic coefficients $a_m \in \mathbb{R}^{p_m}$ and $a_g \in \mathbb{R}^{p_g}$. Find also all open-loop equilibrium configurations q_e , i.e., such that $g(q_e) = 0$.

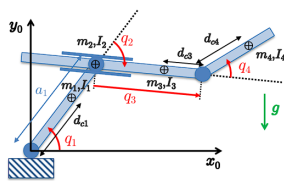


Figure 1: A 4-dof RRPR robot: generalized coordinates, kinematic and dynamic parameters.

$$w_1 = \dot{q}_1 \quad w_2 = \dot{q}_1 + \dot{q}_2 \quad w_3 = \dot{q}_1 + \dot{q}_2 \quad w_4 = \dot{q}_1 + \dot{q}_2 + \dot{q}_4$$

$$p_{c1} = \begin{cases} d_{c1} c_1 \\ d_{c1} s_1 \end{cases} \quad p_{c2} = \begin{cases} z_1 c_1 \\ z_1 s_1 \end{cases} \quad p_{c3} = \begin{cases} z_1 c_1 + (q_3 - d_{c3}) c_{12} \\ z_1 s_1 + (q_3 - d_{c3}) s_{12} \end{cases} \quad p_{c4} = \begin{cases} z_1 c_1 + q_2 c_{12} + d_{c4} c_{124} \\ z_1 s_1 + q_2 s_{12} + d_{c4} s_{124} \end{cases}$$

$$v_{c1} = \begin{cases} -d_{c1} s_1 \dot{q}_1 \\ d_{c1} c_1 \dot{q}_1 \end{cases} \quad T_1 = \frac{1}{2} \dot{q}_1^2 (m_1 d_{c1}^2 + I_1) \quad T_2 = \frac{1}{2} I_2 (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1 \dot{q}_2) + \frac{1}{2} m_2 z_1^2 \dot{q}_1^2$$

$$v_{c3} = \begin{cases} -z_1 s_1 \dot{q}_1 - (q_3 - d_{c3}) s_{12} (\dot{q}_1 + \dot{q}_2) + \dot{q}_3 c_{12} \\ z_1 c_1 \dot{q}_1 + (q_3 - d_{c3}) c_{12} (\dot{q}_1 + \dot{q}_2) + \dot{q}_3 s_{12} \end{cases}$$

$$T_3 = \frac{1}{2} I_3 (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1 \dot{q}_2) + \frac{1}{2} m_3 (z_1^2 \dot{q}_1^2 + (q_3 - d_{c3})^2 (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1 \dot{q}_2) + \dot{q}_3^2 + 2 z_1 s_2 \dot{q}_1 \dot{q}_3 + 2 z_1 (q_3 - d_{c3}) c_2 (\dot{q}_1^2 + \dot{q}_1 \dot{q}_2))$$

$$v_{c4} = \begin{cases} -z_1 s_1 \dot{q}_1 - q_2 s_{12} (\dot{q}_1 + \dot{q}_2) + \dot{q}_2 c_{12} - d_{c4} s_{124} (\dot{q}_1 + \dot{q}_2 + \dot{q}_4) \\ z_1 c_1 \dot{q}_1 + q_2 c_{12} (\dot{q}_1 + \dot{q}_2) + \dot{q}_2 s_{12} + d_{c4} c_{124} (\dot{q}_1 + \dot{q}_2 + \dot{q}_4) \end{cases}$$

$$T_4 = \frac{1}{2} m_4 [z_1^2 \dot{q}_1^2 + q_2^2 (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1 \dot{q}_2) + \dot{q}_2^2 + d_{c4}^2 (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_4^2 + 2\dot{q}_1 \dot{q}_2 + 2\dot{q}_1 \dot{q}_4 + 2\dot{q}_2 \dot{q}_4) + 2 z_1 q_2 c_2 (\dot{q}_1^2 + \dot{q}_1 \dot{q}_2) + 2 z_1 s_2 \dot{q}_1 \dot{q}_2 + 2 z_1 d_{c4} c_{24} (\dot{q}_1^2 + \dot{q}_1 \dot{q}_2 + \dot{q}_1 \dot{q}_4) + 2 q_2 d_{c4} c_4 (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1 \dot{q}_2 + \dot{q}_1 \dot{q}_4 + \dot{q}_2 \dot{q}_4) + 2 d_{c4} c_4 (\dot{q}_1 \dot{q}_2 + \dot{q}_2^2 + \dot{q}_2 \dot{q}_4)] + \frac{1}{2} I_4 (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_4^2 + 2\dot{q}_1 \dot{q}_2 + 2\dot{q}_1 \dot{q}_4 + 2\dot{q}_2 \dot{q}_4)$$

$$M_{11} = m_1 d_{c1}^2 + I_1 + I_2 + m_2 z_1^2 + I_3 + m_3 (z_1^2 + (q_3 - d_{c3})^2 + 2 z_1 (q_3 - d_{c3}) c_2) + I_4 + m_4 (z_1^2 + q_2^2 + d_{c4}^2 + 2 z_1 q_2 c_2 + 2 z_1 d_{c4} c_{24} + 2 q_2 d_{c4} c_4)$$

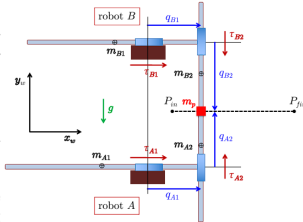
$$M_{12} = I_2 + I_3 + m_3 ((q_3 - d_{c3})^2 + z_1 (q_3 - d_{c3}) c_2) + m_4 (q_2^2 + d_{c4}^2 + z_1 q_2 c_2 + z_1 s_2 + z_1 d_{c4} c_{24} + q_2 d_{c4} c_4 + d_{c4} c_4) + I_4$$

Exercise #2

Consider the situation depicted in Fig. 2. The two Cartesian robots A and B are commanded by motors and transmissions that generate linear forces $\tau_A \in \mathbb{R}^2$ and $\tau_B \in \mathbb{R}^2$ along the axes of their prismatic joints. Each control force component is bounded as

$$|\tau_{A,i}| \leq \tau_{A,max}, \quad i = 1, 2, \quad |\tau_{B,i}| \leq \tau_{B,max}, \quad i = 1, 2.$$

The two robots hold firmly a payload mass m_p and cooperate in moving the mass in minimum time along a horizontal path $p = p(s)$, from point P_{in} to point P_{fin} in a rest-to-rest mode. In addition, it is desired that the sum $H = \|\tau_A\|^2 + \|\tau_B\|^2$ is always minimized instantaneously. Motion occurs in the vertical plane, and we assume that motors are powerful enough to sustain the weight of the respective robot and of the payload. Provide a dynamic model of this cooperating task and determine the optimal time profiles of the four commands τ_A and τ_B .



$$\begin{cases} (m_{A1} + m_{A2}) \ddot{q}_{A1} = \tau_{A1} \\ m_{A2} \ddot{q}_{A2} + m_{A2} g_0 = \tau_{A2} \end{cases} \quad \begin{cases} (m_{B1} + m_{B2}) \ddot{q}_{B1} = \tau_{B1} \\ m_{B2} \ddot{q}_{B2} + m_{B2} g_0 = \tau_{B2} \end{cases}$$

both the robots must hold m_p in an horizontal path hence τ_{A2} and τ_{B2} will be constant.

To minimize the torques norm we have to distribute properly m_p between τ_{A2} and τ_{B2} .

$$\tau_{A2} + \tau_{B2} = m_{A2} \ddot{q}_{A2} + (m_{A2} + \alpha m_p) g_0 + m_{B2} \ddot{q}_{B2} + (m_{B2} + (1-\alpha) m_p) g_0 \quad \text{where } \alpha \in [0, 1]$$

α measure how we distribute m_p . We need the robots to keep

$$\ddot{q}_{A2} = \ddot{q}_{B2} = 0 \quad \text{hence}$$

$$\tau_{A2} + \tau_{B2} = (m_{A2} + \alpha m_p) g_0 + (m_{B2} + (1-\alpha) m_p) g_0$$

$$\frac{1}{g_0} (\tau_A^2 + \tau_B^2) = m_{A2}^2 + \alpha^2 m_p^2 + 2 m_{A2} m_p \alpha + m_{B2}^2 + (1-\alpha)^2 m_p^2 + 2 m_{B2} m_p (1-\alpha) =$$

$$\min_{\alpha} \alpha^2 (2 m_p^2) + \alpha (2 m_{A2} m_p + 2 m_{B2} m_p - 2 \alpha m_p^2) + C \leftarrow \text{const}$$

$$\min_{\alpha} \alpha^2 m_p^2 + \alpha (m_p (m_{A2} + m_{B2} - m_p)) \in \mathcal{S}$$

$$\frac{\partial \mathcal{S}}{\partial \alpha} = 2 \alpha m_p^2 + (m_p (m_{A2} + m_{B2} - m_p)) = 0$$

$$\alpha = - \frac{(m_p (m_{A2} + m_{B2} - m_p))}{2 m_p^2} \Rightarrow \begin{array}{l} \text{this value for} \\ \alpha \text{ minimize} \\ \text{the torque norm} \end{array}$$

Now to perform a rest-to-rest in min. time we compute the max. horizontal acceleration:

$$\begin{cases} (m_{A1} + m_{A2}) \ddot{q}_{A1M} = \tau_{AM} \\ (m_{B1} + m_{B2}) \ddot{q}_{B1M} = \tau_{BM} \end{cases}$$

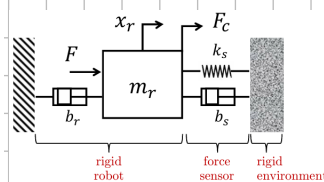
$$\downarrow \quad \downarrow$$

$$\ddot{q}_{A1M} = \frac{\tau_{AM}}{(m_{A1} + m_{A2})} \quad \ddot{q}_{B1M} = \frac{\tau_{BM}}{(m_{B1} + m_{B2})}$$

execute $\tau_{A1} = \tau_{B1}$ to accomplish $\ddot{q}_0$

$$\text{let } \ddot{q}_M = \min \{ \ddot{q}_{A1M}, \ddot{q}_{B1M} \} \Rightarrow \text{we set } \ddot{q}_{id} = \begin{cases} \ddot{q}_M & \text{for } t \in [0, T/2] \\ -\ddot{q}_M & \text{for } t \in [T/2, T] \end{cases}$$

Figure 4 shows a simple 1-dof model of the interaction between a robot of mass $m_r > 0$, commanded by a force F , and a rigid environment, with a force sensor of stiffness $k_s > 0$ measuring the contact force F_c . The two coefficients $b_r > 0$ and $b_s > 0$ represent, respectively, the viscous friction affecting robot motion and the viscous damping of the force sensor. The reference position $x_d = 0$ is when the robot mass is in contact with $F_c = 0$. Provide the dynamic model of this system assuming linearity of all effects. Based only on the measured force, design a control law for F that is able to regulate asymptotically the contact force F_c to a desired constant value $F_d > 0$, despite uncertainty in all model parameters. Provide the associated steady-state values of $F = \bar{F}$ and $x_r = \bar{x}_r$.



$$m\ddot{x} + \underbrace{(b_r + b_s)}_{F_c} \dot{x} + k_s x = F, \quad \text{we want } F_c = k_s x = F_d \Rightarrow x_d = \frac{F_d}{k_s}$$

For the uncertainties

So we choose $F = K_p(x_d - x) + K_I \int (x_d - x) dt \Rightarrow$ we denote $b_r + b_s = b$
where $K_p, K_I > 0$

$$m\ddot{x} + (b_r + b_s)\dot{x} + k_s x = K_p(x_d - x) + K_I \int (x_d - x) dt \quad \text{in the Laplace domain:}$$

$$m s^2 + b s + k_s x + K_p x + K_I \frac{1}{s} x = K_p x_d + \frac{1}{s} K_I x_d$$

$$x (m s^2 + b s + (K_s + K_p) + K_I \frac{1}{s}) = x_d (K_p + K_I \frac{1}{s}) \Rightarrow$$

$$\frac{x}{x_d} = \frac{(K_p + K_I \frac{1}{s})}{(m s^2 + b s + (K_s + K_p) + K_I \frac{1}{s})}$$

Assumption: the system is stable: $\lim_{t \rightarrow \infty} \frac{x}{x_d} = \lim_{s \rightarrow 0} s \frac{(K_p + K_I \frac{1}{s})}{(m s^2 + b s + (K_s + K_p) + K_I \frac{1}{s})} = \frac{K_I}{K_I} = 1$

$\Rightarrow$ if is stable $\lim_{t \rightarrow \infty} x = x_d \Rightarrow F_c = F_d$. the system is stable iff

$m s^3 + b s^2 + (K_s + K_p) s + K_I$ have negative real part roots hence

K_p and K_I must be chosen properly. $\Rightarrow$

$$K_p + K_s > \frac{m K_I}{b} \Rightarrow K_p > \frac{m K_I}{b} - K_s.$$